

| T, °C | 1.5C + 2NF <sub>3(g)</sub> = 1.5CF <sub>4(g)</sub> + N <sub>2(g)</sub> |           |          | C + 4HF <sub>(g)</sub> = CF <sub>4(g)</sub> + 2H <sub>2(g)</sub> |           |          |
|-------|------------------------------------------------------------------------|-----------|----------|------------------------------------------------------------------|-----------|----------|
|       | ΔH                                                                     | ΔS        | ΔG       | ΔH                                                               | ΔS        | ΔG       |
|       | kJ/mol F                                                               | J/K/mol F | kJ/mol F | kJ/mol F                                                         | J/K/mol F | kJ/mol F |
| 100   | -189.4                                                                 | 8.9       | -192.7   | 39.9                                                             | -44.7     | 56.6     |
| 200   | -189.5                                                                 | 8.7       | -193.6   | 40.0                                                             | -44.5     | 61.1     |
| 300   | -189.6                                                                 | 8.4       | -194.5   | 40.2                                                             | -44.1     | 65.5     |
| 400   | -189.8                                                                 | 8.1       | -195.3   | 40.5                                                             | -43.7     | 69.9     |
| 500   | -190.0                                                                 | 7.9       | -196.1   | 40.9                                                             | -43.2     | 74.2     |
| 600   | -190.2                                                                 | 7.6       | -197.0   | 41.3                                                             | -42.7     | 78.5     |
| 700   | -190.4                                                                 | 7.4       | -197.6   | 41.7                                                             | -42.2     | 82.8     |
| 800   | -190.6                                                                 | 7.2       | -198.3   | 42.1                                                             | -41.8     | 87.0     |
| 900   | -190.8                                                                 | 7.0       | -199.1   | 42.6                                                             | -41.4     | 91.1     |
| 1000  | -191.0                                                                 | 6.8       | -199.8   | 43.0                                                             | -41.0     | 95.3     |